Smart Oven Controller Using PIC16F877A

PIC16F877A Embedded Systems Project

Project Overview

This project implements a **Smart Oven Controller** using the **PIC16F877A** microcontroller.

The system allows the user to enter a cooking time, start or pause the countdown, monitor temperature and humidity, detect gas leakage, activate a child lock, and display the system status on a 16x2 LCD.

The controller uses a non-blocking main loop with a Timer0-based system tick for time management.

Main Features

- Time entry using a 4x3 keypad in `MMSS` format.
- Start, pause/resume, reset, and mode buttons.
- Countdown timer controlled by Timer0 interrupt.
- Temperature and humidity monitoring using DHT22.
- Gas leakage detection using an MQ-2 digital output.
- Emergency stop mode when gas is detected.
- Emergency reset requires the gas signal to become safe and the reset button to be held for 3 seconds.
- Child lock using a long press on the `*` key.
- Direct access to the temperature screen using a long press on the `#` key.
- Four LCD screens:
- Main screen
- Temperature and humidity screen
- Safety screen
- System screen
- 10-segment bargraph for countdown progress.
- Green, yellow, and red status LEDs.
- Buzzer patterns for finished and emergency states.
- Button and keypad debouncing.
- Temporary LCD notification messages.

Microcontroller and Software

- **Microcontroller:** PIC16F877A
- **Oscillator Frequency:** 20 MHz
- **Oscillator Mode:** HS
- **Compiler:** MPLAB XC8
- **Development Environment:** MPLAB X IDE
- **Programming Language:** Embedded C

Required Hardware

- PIC16F877A microcontroller
- 20 MHz crystal oscillator
- Two crystal capacitors
- 16x2 LCD
- 4x3 matrix keypad
- DHT22 temperature and humidity sensor

- MQ-2 gas sensor module
- 10-segment LED bargraph
- Green LED
- Yellow LED
- Red LED
- Buzzer
- Start button
- Pause button
- Reset button
- Mode button
- Pull-up and current-limiting resistors
- 5 V regulated power supply
- PIC programmer such as PICkit

Pin Mapping

PORTA

Pin	Connected Device	Description
RA0	Start button	Starts the countdown
RA1	Pause button	Pauses or resumes the countdown
RA2	DHT22 data	Temperature and humidity data line
RA3	MQ-2 gas output	Active-high gas detection input
RA4	Reset button	Resets the system
RA5	Mode button	Changes the LCD screen

PORTB

Pin	Connected Device	Description
RB0	Keypad Row 1	Keypad output
RB1	Keypad Row 2	Keypad output
RB2	Keypad Row 3	Keypad output
RB3	Keypad Row 4	Keypad output
RB4	Keypad Column 1	Keypad input
RB5	Keypad Column 2	Keypad input
RB6	Keypad Column 3	Keypad input
RB7	Buzzer	Alarm output

PORTC

Pin	Connected Device	Description
RC0	LCD RS	LCD register select
RC1	LCD EN	LCD enable
RC2	LCD D4	LCD data
RC3	LCD D5	LCD data
RC4	LCD D6	LCD data

Pin	Connected Device	Description
RC5	LCD D7	LCD data
RC6	Bargraph Segment 1	First bargraph segment
RC7	Bargraph Segment 2	Second bargraph segment

PORTD

Pin	Connected Device	Description
RD0-RD7	Bargraph Segments 3-10	Remaining bargraph segments

PORTE

Pin	Connected Device	Description
RE0	Green LED	Running indicator
RE1	Yellow LED	Paused or finished indicator
RE2	Red LED	Emergency indicator

Keypad Layout

```
1  2  3
4  5  6
7  8  9
*  0  #
```

Keypad Functions

- Numeric keys enter the cooking time in `MMSS` format.
- Long press `*` for 1.2 seconds to enable or disable the child lock.
- Long press `#` for 2 seconds to open or close the temperature screen.

The tens digit of the seconds must be between `0` and `5`.

Example:

```
1234 = 12 minutes and 34 seconds
```

Push Button Functions

Button	Function
Start	Starts the countdown when a valid time is entered
Pause	Pauses or resumes the countdown
Reset	Returns the system to setting mode
Mode	Changes between the available LCD screens

The buttons are active-low and use software debouncing.

System States

The system uses a state machine with the following states:

1. Setting

The user enters the desired cooking time using the keypad.

2. Running

The countdown is active.
The green LED is turned on.

3. Paused

The countdown is temporarily stopped.
The yellow LED is turned on.

4. Finished

The countdown has reached zero.

During the first 5 seconds:

- The LEDs cycle between green, red, and yellow.
- The buzzer switches on and off.
- The complete bargraph flashes.

After 5 seconds:

- The yellow LED remains on.
- The buzzer remains on.

5. Emergency

The system enters emergency mode when the gas sensor output becomes high.

During emergency mode:

- The countdown is stopped.
- The red LED flashes.
- The buzzer generates an emergency alarm pattern.
- The bargraph creates a scanning animation.
- The LCD displays an emergency warning.

To leave emergency mode:

1. The gas sensor signal must return to the safe state.
2. The reset button must be held for 3 seconds.

LCD Screens

Main Screen

Displays:

- Remaining time
- Current system state

Example:

```
TIME:   12:34
STATE:  RUN
```

Temperature Screen

Displays:

- Temperature in degrees Celsius
- Relative humidity percentage

If the DHT22 reading fails, the LCD displays `---`.

Safety Screen

Displays:

- Gas sensor state
- Alarm state

System Screen

Displays:

- Child lock status
- System readiness

Timer0 Operation

Timer0 generates an interrupt approximately every 1 millisecond.

Configuration:

- Oscillator frequency: 20 MHz
- Instruction clock: `Fosc / 4 = 5 MHz`
- Prescaler: `1:32`
- Timer tick: approximately `6.4 us`
- Timer preload value: `100`
- Overflow period: approximately `1 ms`

The interrupt service routine:

1. Reloads Timer0.
2. Increments the global millisecond counter.
3. Counts 1000 milliseconds to generate one second.
4. Decreases the remaining time while the system is running.
5. Changes the system state to `ST_FINISHED` when the remaining time reaches zero.

DHT22 Operation

The DHT22 is connected to RA2.

The program:

1. Temporarily configures RA2 as an output.
2. Sends a low start signal.
3. Releases the line and changes RA2 to an input.
4. Reads 40 bits from the sensor.
5. Stores the received data in five bytes.
6. Checks the checksum.
7. Converts the raw values into temperature and humidity.

Interrupts are temporarily disabled during communication because the DHT22 protocol requires accurate timing.

The temperature screen refreshes approximately every 2.5 seconds.

Bargraph Operation

The 10-segment bargraph shows the remaining cooking time as ten levels.

The number of active segments is calculated using:

```
Active segments = Remaining time x 10 / Total time
```

Special patterns:

- Emergency: one segment scans from side to side.

- Finished: all segments flash.

Project Operation

1. Power on the circuit.
2. Wait for the welcome message.
3. Enter four digits using the keypad.
4. Press the Start button.
5. Use Pause to stop or resume the countdown.
6. Use Mode to change the LCD screen.
7. Use Reset to clear the entered time and return to setting mode.
8. Hold `*` to activate or deactivate the child lock.
9. Hold `#` to open or close the temperature screen.

Building the Project

1. Open MPLAB X IDE.
2. Create a standalone project for the PIC16F877A.
3. Select the connected PIC programmer.
4. Select the XC8 compiler.
5. Add the source code file to the project.
6. Make sure the oscillator frequency is set to 20 MHz.
7. Build the project.
8. Connect the PIC programmer.
9. Program the generated HEX file into the PIC16F877A.

Configuration Bits

```
#pragma config FOSC  = HS
#pragma config WDTE  = OFF
#pragma config PWRTE = ON
#pragma config BOREN = ON
#pragma config LVP   = OFF
#pragma config CPD   = OFF
#pragma config WRT   = OFF
#pragma config CP    = OFF
```

Important Notes

- The LCD operates in 4-bit mode.
- The DHT22 data line requires a pull-up resistor.
- The keypad columns use pull-up logic.
- The buttons are active-low.
- The MQ-2 input is treated as active-high.
- PORTA analog functions and comparators are disabled in software.
- The buzzer and keypad share PORTB, so PORTB is updated through a common function.
- The countdown is controlled by Timer0 and does not depend on long blocking delays.
- Ensure that the circuit uses a stable 5 V supply and a common ground.